ThoughtFlow: A Sparse-Cache Falsification Study for Reasoning-Trace Retention

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Abstract

The ThoughtFlow-FP8 project tests whether training-free sparse KV retention can preserve useful reasoning context better than matched-budget KV-retention proxies. The current contribution is diagnostic, not a positive method, and no real FP8, CUDA, latency, or throughput result is claimed. Synthetic traces support the intuition that phase markers and anchors can be protected, but real retained-context quality and sparse-cache gates rule out further tuning of the original anchor/recent/phase/math family on the available traces. We pre-registered one successor signal, recurrence-distance utility (`rdu_topk`), which ranks prefix tokens by delayed self-attention reuse across lag buckets and uses no token labels, recent reserve, or continuation loss. On the frozen 74-trace distilgpt2 sparse-cache gate, `rdu_topk` initially reaches NLL 3.779 versus ThinKV-like 3.900 and R-KV-like 3.939. Reproduction gates demote it: a same-slice rerun reproduces the cached gate exactly, but an alternate-surface check fails same-family separation by 0.006 NLL, and a larger independent saved-trace check fails cross-family separation, with R-KV-like at NLL 3.981 versus `rdu_topk` at 4.014 on 89 scored traces. The anchor/phase int8 quantization primitive passes Triton interpreter correctness tests against its CPU reference. The paper therefore reports a sparse-cache falsification ladder and a stopped method branch.

1 Policy Family

The original policy protects anchors and phase labels. The first successor adds a small recent-token reserve. The second successor protects anchors, reserves half the retained budget for recent tokens, and lets phase markers compete with math-state and high-importance reasoning tokens under a saliency bonus. Those policies are now stopped on the current saved traces. The pre-registered `rdu_topk` successor is also demoted to a diagnostic after stricter reproduction; it remains useful only for explaining which sparse-cache surfaces are fragile.

Policy	Keep rate	NLL	Δ NLL vs full	PPL
full context	1.000	2.101	0.000	9.7
R-KV-like	0.210	3.419	1.319	38.7
ThinKV-like	0.210	3.583	1.482	44.6
LongFlow-like	0.210	3.961	1.861	74.2
ThoughtFlow	0.210	3.961	1.861	74.2
ThoughtFlow-recent	0.210	3.562	1.461	44.2
ThoughtFlow-saliency-recent	0.210	3.434	1.333	38.1
Held-out R-KV-like	0.211	3.482	–	43.7
Held-out sweep best	0.211	3.480	–	43.7

Table 1: Retained-context continuation proxy on distilgpt2 traces. Lower NLL is better. Full context is not a matched-budget baseline. The held-out sweep row is selected on 12 train traces and reported on 12 held-out traces, so its NLL is not directly averaged with the all-trace rows.

2 Hidden/KV Telemetry

We then ran a Mac-local telemetry gate using real distilgpt2 tensors rather than only text markers. The probe ranks retained tokens by attention received, final-hidden norm, key norm, value norm, or combined KV norm. This is still not sparse-KV decoding, but it checks whether the interpretable phase policy is merely preserving labels that hidden/KV saliency would not select.

Policy	Keep rate	Anchor	Phase	Math-state
ThoughtFlow	0.207	1.000	1.000	0.918
LongFlow-like	0.207	1.000	1.000	0.918
ThinKV-like	0.207	1.000	0.948	0.848
`value_norm_topk`	0.207	0.000	0.492	0.845
train-selected sweep	0.207	1.000	0.430	0.956

Table 2: Hidden/KV saliency telemetry on 24 distilgpt2 traces. The strongest real-saliency proxy by phase+math recall is `value_norm_topk`. ThoughtFlow beats it on phase recall, but LongFlow-like ties ThoughtFlow under the local importance heuristic.

Paired against `value_norm_topk`, ThoughtFlow's phase-recall delta is +0.508 with 95% CI [+0.436, +0.579]. The math-state delta is only +0.073 with 95% CI [-0.078, +0.223]. This rules out a reviewer-facing claim based on phase recall alone: it may be marker preservation rather than useful cache retention.

3 CPU Sparse-Cache Quality Gate

Finally, we ran a CPU-only sparse-cache probe. The model processes the full prefix once, each policy prunes the returned KV cache to the same 0.20 budget, and the continuation is scored from the pruned cache. This is not a Triton, CUDA, FP8, latency, or throughput result, but it is stronger than retained-text NLL because the model actually receives a sparse cache.

Policy	Keep rate	NLL	Paired Δ vs R-KV-like
full cache	1.000	2.142	-1.296 [-1.533,-1.066]
ThoughtFlow-saliency-recent	0.210	3.372	-0.067 [-0.151,+0.011]
ThinKV-like	0.210	3.389	-0.049 [-0.192,+0.077]
ThoughtFlow-recent	0.210	3.399	-0.040 [-0.169,+0.074]
ThoughtFlow sweep best	0.210	3.436	-0.002 [-0.007,+0.000]
R-KV-like	0.210	3.438	0.000
LongFlow-like	0.210	3.588	+0.150 [-0.016,+0.300]
ThoughtFlow	0.210	3.588	+0.150 [-0.006,+0.297]

Table 3: CPU sparse-KV continuation quality on 24 distilgpt2 traces. Negative paired delta is better than R-KV-like. ThoughtFlow-saliency-recent is best in mean NLL, but it clears the strongest non-ThoughtFlow row by only 0.017 NLL and remains below the pre-registered 0.03 promotion margin.

A bounded sparse-cache sweep over 24 nearby anchor/recent/phase/math configurations selects `tf_sparse_r0.55_p0.05_m0.12_a2` on the 12 train traces. On the 12 held-out traces it gets NLL 3.340 versus ThinKV-like 3.385 and R-KV-like 3.420. This clears the mean 0.03 margin, and the paired delta versus R-KV-like is -0.080 with 95% CI [-0.152,-0.014]. However, the paired delta versus ThinKV-like is -0.045 with 95% CI [-0.226,+0.182]. The fixed ThoughtFlow-saliency-recent incumbent has lower held-out mean NLL, 3.304, but its paired CIs also cross zero. This is a promising falsification candidate, not a positive result.

We then froze `ThoughtFlow-saliency-recent` and `tf_sparse_r0.55_p0.05_m0.12_a2` and ran a larger 74-trace sparse cache slice with no retuning. ThinKV-like is the best compressed row among the stopped family comparison at NLL 3.900. The frozen sparse ThoughtFlow row gets 3.908, with paired delta -0.031 versus R-KV-like and 95% CI [-0.078,+0.020], but +0.008 versus ThinKV-like with 95% CI [-0.060,+0.085]. ThoughtFlow-saliency-recent gets 3.920. The larger no-retuning gate therefore stops this policy family.

After that stop decision, we pre-registered exactly one successor, `rdu_topk`. It computes recurrence-distance utility from full-prefix prefill attentions using lag buckets [8,15], [16,31], [32,63], and [64,∞], then keeps the top budget by utility with ties broken by lower position. It does not use continuation tokens, trace labels, recent reserves, anchors, phase markers, or math-state bonuses.

4 Evidence Ladder

Gate	Readout	Decision
Synthetic retention	Anchor/phase labels can be protected at matched keep rate.	Intuition only.
Retained-text NLL	Original ThoughtFlow rows do not beat the strongest proxy.	Weakened.
Hidden/KV telemetry	Phase recall improves, but math-state impact is not stable and LongFlow-like ties the heuristic.	Diagnostic.
CPU sparse-cache probe	A tuned sparse ThoughtFlow row is promising in mean but not robust against ThinKV-like uncertainty.	Not promoted.
Frozen 74-trace gate	`rdu_topk` beats ThinKV-like and R-KV-like on the first frozen surface.	Candidate only.
Same-slice rerun	Cached result reproduces exactly on the deterministic Mac stack.	Repro bookkeeping.
Alternate surface	Cross-family rows still lose, but a stopped same-family row beats `rdu_topk` by 0.006 NLL.	Weakened.
Independent saved traces	R-KV-like is best compressed at NLL 3.981; `rdu_topk` reaches 4.014.	Killed.

Table 4: ThoughtFlow-FP8 evidence ladder. The positive-looking gate is shown as one step that failed to reproduce, not as the headline result.

Policy	Traces	Keep rate	NLL	Paired Δ vs ThinKV-like
full cache	74	1.000	2.848	-1.052 [-1.199,-0.919]
`rdu_topk`	74	0.213	3.779	-0.121 [-0.211,-0.037]
ThinKV-like	74	0.213	3.900	0.000
frozen sparse ThoughtFlow	74	0.213	3.908	+0.008 [-0.060,+0.085]
ThoughtFlow-saliency-recent	74	0.213	3.920	+0.020 [-0.030,+0.074]
R-KV-like	74	0.214	3.939	+0.039 [-0.015,+0.099]
LongFlow-like	74	0.213	4.158	+0.258 [+0.178,+0.337]

Table 5: One-shot frozen sparse-cache successor evaluation. `rdu_topk` also beats R-KV-like by -0.160 NLL with 95% CI [-0.264,-0.050], clearing the pre-registered promotion rule.

Cached split diagnostic. We then reused the same 74-trace rows without rerunning the model or changing the `rdu_topk` rule. Even/odd and first-half/second-half partitions all keep `rdu_topk` as the best compressed row and preserve at least 0.03 mean NLL margin versus both R-KV-like and ThinKV-like. However, only 2/4 half-size partitions keep both paired confidence intervals below zero. This supports the current promotion decision but does not replace an independent larger or seed-repeated reproduction.

Measured no-retuning rerun. We also reran the frozen probe on the same 74 traces and wrote a separate measured artifact rather than overwriting the cached gate. On the current deterministic Mac stack, all policy NLLs match the cached promoted gate exactly: `rdu_topk` remains best compressed at NLL 3.779, with zero measured minus cached NLL drift. Strict separation is preserved: measured margins versus `rdu_topk` are +0.141 and +0.129 NLL for the stopped ThoughtFlow rows, and +0.160, +0.121, and +0.379 for R-KV-like, ThinKV-like, and LongFlow-like. The measured per-trace compressed oracle reaches NLL 3.634, leaving `rdu_topk` 0.145 NLL above that oracle and 0.931 NLL above full cache, with a 0.419 oracle-hit rate. This is useful reproduction bookkeeping, not a new independent slice.

Measured alternate-surface check. We then changed only the measurement surface, using `max_length=112` and `continuation_tokens=32`, while preserving the same `rdu_topk` rule, matched 0.20 keep fraction, cached-versus-measured labels, same-family/cross-family reporting, and oracle/headroom diagnostics. This check is not a clean reproduction. `rdu_topk` remains strong against cross-family rows, reaching NLL 3.594 versus R-KV-like 3.681 and ThinKV-like 3.851, with paired deltas -0.087 [-0.139,-0.028] and -0.256 [-0.465,-0.086]. However, `tf_sparse_r0.55_p0.05_m0.12_a2`, a stopped same-family sparse row, reaches NLL 3.588 and beats `rdu_topk` by 0.006. The measured per-trace compressed oracle reaches NLL 3.460, leaving `rdu_topk` 0.135 above oracle and 0.847 above full cache, with a 0.438 oracle-hit rate.

Independent saved-trace reproduction check. The stronger no-retuning check changes the saved trace inputs to a 96-row chat SVAMP surface while keeping the same `rdu_topk` rule and 0.20 keep fraction. It scores 89 traces after token-length filtering. This does not reproduce the positive gate. R-KV-like is now the best compressed row at NLL 3.981, while `rdu_topk` reaches 4.014. The paired delta versus R-KV-like is +0.032 with 95% CI [-0.071,+0.137], and ThinKV-like is only 0.030 NLL worse than `rdu_topk` with a confidence interval crossing zero. Same-family stopped rows are worse than `rdu_topk`, but only by 0.006 and 0.010 NLL. The compressed per-trace oracle reaches NLL 3.754, leaving `rdu_topk` 0.260 above oracle and 1.083 above full cache. This rules out claiming `rdu_topk` as a robust positive method on the available surfaces.

A coarse trace-level decomposition explains where the reproduction fails. On long-prefix/high-RDU-density rows, `rdu_topk` is +0.213 NLL worse than R-KV-like; on short-prefix/low-density rows it is -0.049 better. When R-KV-like is the per-trace compressed oracle, `rdu_topk` is +0.498 NLL worse; when the stopped sparse ThoughtFlow row is oracle, it is +0.174 worse. The failure is therefore not a uniform collapse, but the large-regression buckets are exactly the buckets a robust policy would need to handle before any long-context claim.

5 Decision

The branch is now demoted to a diagnostic. The stopped anchor/recent/phase/math policy family remains ruled out as a tunable branch on this trace set, the alternate-surface check fails same-family separation, and the independent saved-trace check fails cross-family separation. A paper-ready claim now requires a genuinely new pre-registered utility signal evaluated once on a fresh/larger frozen surface with strict same-family, cross-family, and oracle/headroom reporting. The Triton interpreter gate is no longer the blocker for the quantization scaffold; method evidence is.

6 Related Work and Scope

This draft intentionally does not claim a new state-of-the-art KV-compression method. ThinKV, LongFlow, R-KV/R-KVHash-style retention, LazyEviction/ForesightKV-style future-use ideas, PM-KVQ, and KVQuant already occupy much of the sparse-cache and quantized-cache design space. Our contribution is narrower: a Mac-local falsification ladder that shows how a plausible interpretable retention signal can clear one frozen surface and then fail stricter same-family and cross-family checks. The project name retains "FP8" because that was the original systems target, but the current artifacts only validate CPU sparse-cache scoring and an int8/Triton-interpreter reference primitive; they do not measure FP8 numerical drift or native FP8 serving.

7 Reproducibility

All reported gates use tracked scripts under `experimental/thoughtflow_fp8/phase2/` and the repo-local `./venv_arm64`. The key reproduction artifacts are the frozen sparse-cache probe, same-slice rerun, alternate-surface rerun, and independent saved-trace rerun listed below. Promotion criteria, trace filtering, paired intervals, oracle/headroom diagnostics, and no-retuning decisions are written into the corresponding Markdown and JSON artifacts. The stable local correctness command is:

```
TRITON_CPU_BACKEND=1 TRITON_INTERPRET=1 ./venv_arm64/bin/python
-m pytest experimental/thoughtflow_fp8/phase2/tests
experimental/thoughtflow_fp8/phase4/tests -rs
```

8 Artifacts

- `experimental/thoughtflow_fp8/phase2/perplexity_impact_proxy.md`
- `experimental/thoughtflow_fp8/phase2/policy_sweep.md`
- `experimental/thoughtflow_fp8/phase2/hidden_saliency_retention_probe.md`
- `experimental/thoughtflow_fp8/phase2/kv_drop_quality_probe.md`
- `experimental/thoughtflow_fp8/phase2/frozen_sparse_cache_probe.md`
- `experimental/thoughtflow_fp8/phase2/rdu_robustness_diagnostic.md`
- `experimental/thoughtflow_fp8/phase2/rdu_no_retune_reproduction_check.md`
- `experimental/thoughtflow_fp8/phase2/rdu_alt_surface_reproduction_check.md`
- `experimental/thoughtflow_fp8/phase2/rdu_independent_trace_reproduction_check.md`
- `experimental/thoughtflow_fp8/phase2/real_trace_retention_sweep.md`
- `experimental/triton_cpu_source_install_20260506.md`
- `experimental/thoughtflow_fp8/paper/colm_workshop_shell.md`